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PR MA1101 MATEMATIKA I

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Jawaban

1. $\lim\limits_{x\to 1} \dfrac{\sqrt{2x}\,|x-1|}{x-1}$ = Tidak ada nilai limitnya.

$x > 1 \Rightarrow |x-1| = x-1$

Bukti. Tinjau $\lim\limits_{x\to 1^+} \dfrac{\sqrt{2x}\,|x-1|}{x-1} = \dfrac{\sqrt{2x}\,(x-1)}{x-1}$

$= \sqrt{2x}$

$= \sqrt{2 \lim\limits_{x\to 1^+} x}$

$= \sqrt{2}$

$x < 1 \Rightarrow |x-1| = -(x-1)$

Tinjau $\lim\limits_{x\to 1^-} \dfrac{\sqrt{2x}\,|x-1|}{x-1} = \dfrac{\sqrt{2x}\,(-(x-1))}{x-1}$

$= -\sqrt{2x}$

$= -\sqrt{2}$

Karena $\lim\limits_{x\to 1^+} \dfrac{\sqrt{2x}\,|x-1|}{x-1} \neq \lim\limits_{x\to 1^-} \dfrac{\sqrt{2x}\,|x-1|}{x-1}$,

nilai limitnya tidak ada

2. $\lim\limits_{x\to 0} \dfrac{\tan^2 3x}{2x} = \lim\limits_{x\to 0} \dfrac{\tan 3x \cdot \tan 3x}{2x}$

$= \lim\limits_{x\to 0} \dfrac{\sin 3x}{\cos 3x} \cdot \dfrac{\sin 3x}{\cos 3x} \cdot \dfrac{1}{2x}$

$= \lim\limits_{x\to 0} \dfrac{\sin 3x}{2x} \cdot \dfrac{3x}{3x} \cdot \dfrac{\sin 3x}{1} \cdot \dfrac{1}{\cos 3x} \cdot \dfrac{1}{\cos 3x}$

$= \lim\limits_{x\to 0} \dfrac{\sin 3x}{3x} \cdot \dfrac{3x}{2x} \cdot \sin 3x \cdot \dfrac{1}{1} \cdot \dfrac{1}{1}$

$= \lim\limits_{x\to 0} 1 \cdot \dfrac{3}{2} \cdot 0 = \boxed{0}$

$\tan x \approx x$
$\Rightarrow \dfrac{(3x)^2}{2x}$
$= \dfrac{9x^2}{2x}$
$= \dfrac{9}{2}x$
$= 0$

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a. $\lim\limits_{x \to -\infty} \dfrac{\sqrt{1+4x^6}}{2-x^3} = \lim\limits_{x \to -\infty} \dfrac{\sqrt{1+4x^6} \cdot \frac{1}{x^3}}{(2-x^3) \cdot \frac{1}{x^3}}$

$= \lim\limits_{x \to -\infty} \dfrac{\sqrt{\frac{1+4x^6}{x^6}}}{\frac{2}{x^3}-1}$

$= \lim\limits_{x \to -\infty} \dfrac{\sqrt{x^{-6}+4}}{2x^{-3}-1} = \dfrac{\sqrt{0+4}}{2(0)-1}$

$= -\sqrt{\frac{4}{1}} = \boxed{-2}$

b. $\lim\limits_{x \to \infty} \sqrt{9x^2+x} - 3x$

$= \lim\limits_{x \to \infty} \sqrt{9x^2+x} - \sqrt{9x^2}$

$= \lim\limits_{x \to \infty} \dfrac{\sqrt{9x^2+x} - \sqrt{9x^2}}{1} \cdot \dfrac{(\sqrt{9x^2+x} + \sqrt{9x^2})}{(\sqrt{9x^2+x} + \sqrt{9x^2})}$

$= \lim\limits_{x \to \infty} \dfrac{9x^2+x-(9x^2)}{\sqrt{9x^2+x} + \sqrt{9x^2}}$

$= \lim\limits_{x \to \infty} \dfrac{x}{\sqrt{9x^2+x} + \sqrt{9x^2}} \cdot \dfrac{\frac{1}{x}}{\frac{1}{x}}$

$= \lim\limits_{x \to \infty} \dfrac{1}{\sqrt{9+\frac{1}{x}} + \sqrt{9}}$

$= \dfrac{1}{\sqrt{9+0} + \sqrt{9}} = \dfrac{1}{3+3}$

$= \boxed{\dfrac{1}{6}}$

c. $\lim\limits_{x \to -\infty} \dfrac{1+x^6}{x^4+1} = \lim\limits_{x \to -\infty} \dfrac{(1+x^6)(x^{-4})}{(x^4+1)(x^{-4})}$

$= \lim\limits_{x \to -\infty} \dfrac{x^{-4}+x^2}{1+\frac{1}{x^{4}}}$

$= (-\infty)^2$

$= \infty^2 = \boxed{\infty}$ (divergen)

(part seru)

Mari kita buktikan bahwa $\lim\limits_{x \to -\infty} \frac{1+x^6}{x^4+1} \to \infty$

menggunakan definisi limit untuk $-\infty \overset{\text{mendekati}}{\Rightarrow} \infty$

Bukti. Misalkan $\exists N > 0 \ni \exists M \; x < M, \; \frac{1+x^6}{x^4+1} > N$

Ambil $M = \min(-1, -\sqrt{2N})$

karena $x < M$, $x < -1 \wedge x < -\sqrt{2N}$

karena $x < -1$, $x^4 > 1 \Rightarrow x^4 + 1 < 2x^4$

$1 + x^6 > x^6 \Rightarrow \frac{1+x^6}{x^4+1} > \frac{x^6}{x^4+1} > \frac{x^6}{2x^4} > \frac{x^2}{2} > N$

Maka, $\exists N > 0, \; \frac{1+x^6}{x^4+1} > N$

Q.E.D.

d. $\lim\limits_{x \to \infty} \frac{\sin x}{x} = \boxed{0}$

$\sqsubset$ Squeeze theorem/sandwich

Bukti 1. Perhatikan bahwa $-1 \leq \sin x \leq 1$

$\Rightarrow \lim\limits_{x \to \infty} -\frac{1}{x} \leq \frac{\sin x}{x} \leq \frac{1}{x}$

$\Rightarrow 0 \leq \frac{\sin x}{x} \leq 0$

$\Rightarrow \lim\limits_{x \to \infty} \boxed{\frac{\sin x}{x} = 0}$ ■

Bukti 2 Menggunakan definisi limit untuk tak hingga

$\forall \varepsilon > 0, \; \exists M < x \ni \left| \frac{\sin x}{x} - 0 \right| < \varepsilon$

Ambil $M = \varepsilon^{-1}$, maka karena $|\sin x| \leq 1$,

$\left| \frac{\sin x}{x} \right| \leq \frac{1}{|x|} \leq \frac{1}{x} < \varepsilon$ ■

$\sqsubset \; x > M > 0 \Rightarrow |x| = x$

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4. a $\lim_{x \to 2^-} \frac{x^2}{4-x^2} = \boxed{\infty}$

Bukti 1 Substitusi x dengan $x = 2-h$ dengan $h>0$, maka ekuivalen dengan

$$\lim_{h \to 0^+} \frac{(2-h)^2}{4-(2-h)^2} = \frac{4+h^2-4h}{4-(4+h^2-4h)}$$
$$= \frac{4+h^2-4h}{4h-h^2} \cdot \frac{\frac{1}{h^2}}{\frac{1}{h^2}}$$
$$= \frac{\frac{4}{h^2}+1-\frac{4}{h}}{\frac{4}{h}-1}$$

Substitusi $h=0$

$$\Rightarrow \frac{4+(0)^2-4(0)}{4(0)-0^2} = \frac{4}{0^+} = \boxed{\infty}$$

Bukti 2 Menggunakan definisi limit, maka

$\forall N>0, \exists \delta>0 \ni 2-\delta<x<2 \Rightarrow \frac{x^2}{4-x^2} > N$

Ambil $\delta = \min(1, \frac{1}{4N})$, maka

$2-\delta<x<2 \Rightarrow x>1 \wedge 2-x<\frac{1}{4N} \ni$

$$\frac{x^2}{4-x^2} = \frac{x^2}{(2-x)(2+x)} > \frac{1}{4(2-x)} > \frac{1}{4 \cdot \frac{1}{4N}} = N$$

Q.E.D

b. $\lim_{t \to \pi^-} \cot t = \boxed{-\infty}$

Bukti. $\forall N<0, \exists \delta>0 \ni \pi-\delta<t<\pi, \cot(t) < N$

Perhatikan bahwa $t>t_0 \Rightarrow \cot(t) < \cot(t_0)$

Ambil $\delta = \pi - \operatorname{arccot}(N)$. Karena cot kontinu, arccot juga kontinu sehingga

$t>\pi-\delta \Rightarrow t>\pi-(\pi-\operatorname{arccot} N)$

$\Rightarrow t > \operatorname{arccot} N$

$\Rightarrow \cot t < N$. ∎

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5. Asimtot datar/tegak dari $f(x) = \frac{1+x^4}{x^2-x^4}$

Asimtot tegak = pembilang nol
datar = lim ∞ atau $-\infty$

$f(x) = \frac{1+x^4}{x^2-x^4} = \frac{1+x^4}{x^2(1-x^2)}$, penyebut nol ketika $x^2(1-x^2) = 0$
$\Rightarrow x \in \{0, -1, 1\}$

$\lim_{x \to -\infty} \frac{1+x^4}{x^2-x^4} = \underbrace{\lim_{x \to \infty} \frac{1+x^4}{x^2-x^4}}_{}$ $= -1$ $\because$ fungsi genap, $f(x) = f(-x)$

Dengan demikian, ada

$\boxed{\text{3 asimtot tegak pada } x=0,\ x=-1, \text{ dan } x=1;\ \text{1 asimtot tegak pada } y=-1}$

6. $f(x) = \frac{x^2+3x-10}{x^2-5x+6} = \frac{(x+5)(x-2)}{(x-3)(x-2)}$

Ketidakkontinunya terdapat pada $x=3$ atau $x=2$ karena pembuat penyebut nol dengan $x=3$ takterhapuskan sementara $x=2$ terhapuskan karena

$$\lim_{x \to 2} \frac{(x+5)(x-2)}{(x-3)(x-2)} = \lim_{x \to 2} \frac{x+5}{x-3} = \frac{7}{-1} = \boxed{-7}$$

$x=3$ takterhapuskan karena berupa asimtot tegak dan nilai

$$\lim_{x \to 3} \frac{(x+5)(x-2)}{(x-3)(x-2)}\ \boxed{\text{tidak ada}}$$

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7. $f(x) = \begin{cases} cx^2 + 2x, & x < 2 \\ x^3 - cx & x \geq 2 \end{cases}$

Agar kontinu, $\lim_{x \to 2} f(x)$ harus ada, maka

$\lim_{x \to 2^-} f(x) = \lim_{x \to 2^+} f(x)$

$\Rightarrow \lim_{x \to 2^-} cx^2 + 2x = \lim_{x \to 2^+} x^3 - cx$

$\Rightarrow 4c + 2c = 8 - 2c$

$\Rightarrow 8c = 8$

$\Rightarrow \boxed{c = 1}$

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8. Bukti bahwa $\exists x \in \mathbb{R} \ni \sqrt{x-5} = \frac{1}{x+3}$ dapat menggunakan

IVT atau Teorema Nilai Tengah / Antara

Konstruksi sebuah fungsi $h(x) = \sqrt{x-5} - \frac{1}{x+3}$

$h(5) = \sqrt{5-5} - \frac{1}{5+3} = -\frac{1}{8} < 0$

$h(6) = \sqrt{6-5} - \frac{1}{6+3} = 1 - \frac{1}{9} = \frac{8}{9} > 0$

Karena h kontinu pada $x > 5$, $\exists c \ni -\frac{1}{8} < 0 < \frac{8}{9}$

dan karena $0 \in [-\frac{1}{8}, \frac{8}{9}]$, maka

$5 < c < 6$. $h(5) < h(c) < h(6)$

Dengan demikian $\boxed{\exists c \in (5, 6) \ni h(c) = 0 \text{ karena } 0 \in (-\tfrac{1}{8}, \tfrac{8}{9})}$

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